

VibeGuard: Local Sourcing Action Guide & Rig Assembly Manual

Scope: Group 3 (Passive Electronic Components) & Group 4 (Mechanical Test Rig Assembly)

Target Audience: Amith Krishna Das (Hardware & Rig Lead) & VibeGuard Integration Team

Institution: Jyothi Engineering College (Autonomous), Cheruthuruthy, Kerala

1. Complete Bill of Materials (Group 3 & Group 4)

Item ID	Component Name	Technical Specification / Code	Qty	Target Sourcing Location	Est. Cost
G3-1	Electrolytic Bulk Capacitor	100 μ F, 16V or 25V (Radial leads, low ESR)	2 pcs	College ECE/Hardware Lab OR Local TV Repair Shop	₹0 – ₹5
G3-2	Ceramic Bypass Capacitor	100 nF (0.1 μ F), marked "104" (Disc or MLCC)	2 pcs	College ECE/Hardware Lab OR Local Electronics Shop	₹0 – ₹2
G3-3	Flyback Clamp Diode	1N4007 (1A, 1000V) or 1N5819 (1A, 40V Schottky)	1 pc	College ECE/Hardware Lab OR Local Electronics Shop	₹0 – ₹2
G3-4	Current-Limiting Resistors	220 Ω , ¼ Watt (Color code: Red-Red-Brown-Gold)	2 pcs	College ECE/Hardware Lab drawer	₹0 – ₹2

Item ID	Component Name	Technical Specification / Code	Qty	Target Sourcing Location	Est. Cost
G4-1	Rig Baseplate	Flat wood, plywood, MDF, or 8–10 mm acrylic (\$\approx 15 \times 15 \text{ cm}\$)	1 pc	College Carpentry / Mech Workshop scrap bin	₹0
G4-2	Eccentric Weight Bolt	M3 × 15 mm (or M3 × 20 mm) machine screw / hex socket bolt	1 pc	College Mech Lab OR Local Nut-and-Bolt Shop	₹1 – ₹2
G4-3	Metal Washers	M3 Flat Steel Washers (Inner dia: 3.2 mm, Outer dia: 7–9 mm)	4 pcs	College Mech Lab OR Local Nut-and-Bolt Shop	₹2
G4-4	Nylon Locknut	M3 Nyloc Nut (Nut with internal nylon ring to prevent loosening)	1 pc	College Mech Lab OR Local Nut-and-Bolt Shop	₹1 – ₹2
G4-5	Motor Wood Screws	M2.5 or M3 wood screws (length 10–12 mm) to mount U-bracket	2 pcs	College Mech Lab OR Local Hardware Shop	₹2
G4-6	3 mm Shaft Coupling / Rotor Arm	3 mm bore hub with set-screw OR 3D-Printed Eccentric Rotor Arm	1 pc	College FabLab / 3D Printer (Recommended) OR Hobby Store	₹0 – ₹10
G4-7	Transparent Safety Guard	Clear plastic enclosure, acrylic box, or transparent	1 pc	Local household shop / Lab scrap plastic	₹0 –

Item ID	Component Name	Technical Specification / Code	Qty	Target Sourcing Location	Est. Cost
		cup ($\approx 8 \times 8 \text{ cm}$)	c	container	₹10

2. Visual Reference: The Assembled Mechanical Rig

[!VibeGuard Mechanical Test Rig Baseplate](#)

Key features shown: (1) Rigid wooden baseplate, (2) Metal N20 mounting bracket bolted to wood, (3) 3 mm shaft hub holding an off-center M3 bolt and washers, (4) ADXL345 sensor bolted rigidly beside the motor, sharing structural vibration.

3. Local Sourcing Action Playbook: Where, Whom & What to Ask

Location A: College Electronics / Hardware Labs (ECE & CSE Departments)

- **Target Components:** 100 μF capacitor, 100 nF ("104") ceramic capacitor, 1N4007 diode, 220 Ω resistors.
- **Whom to Ask:** The Lab Assistant, Technical Officer, or Lab in-charge of the Basic Electronics Lab or Microcontroller Lab.
- **What to Say (Dialogue Script):**

> "Good morning Sir/Madam, we are working on our PBCST504 Microcontroller semester project (VibeGuard vibration monitor). We need a few standard basic passives from the component store if available:

> 1. One or two **100 μF (16V or 25V) electrolytic capacitors** for power filtering.

> 2. Two **100 nF ceramic capacitors** (the small disc ones marked '104').

> 3. One **1N4007 diode** for motor flyback protection.

> 4. Two **220 Ω resistors** for our RGB LED.

> Could you please issue these from the lab inventory drawer?"

- **If unavailable at college:** Any local TV/electronics repair shop or consumer electronics service center in Cheruthuruthy or Shoranur will hand you all four parts from their workbench for ₹10 cash.
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Location B: College Mechanical Workshop & Carpentry Lab

- **Target Components:** The Wooden Baseplate ($\approx 15 \times 15 \text{ cm}$), M3 wood mounting screws, and scrap acrylic.
- **Whom to Ask:** The Carpentry Workshop Instructor or Mechanical Workshop Superintendent.
- **What to Say (Dialogue Script):**

> "Sir, we are engineering an embedded machine monitoring rig for our project. We need a small, sturdy flat baseboard to mount a miniature motor and sensor.

> Do you have a small offcut scrap piece of **plywood, hardwood, or 8–10 mm thick acrylic sheet**, roughly **$15 \times 15 \text{ cm}$ (or 6 inches square)** from the scrap bin? Also, if possible, 2 small wood screws to fasten a small metal bracket?"

- **Key Spec:** The wood must be flat, dense, and at least 8 mm to 12 mm thick so that vibration from the motor travels through the wood into the sensor without flexing or warping.
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Location C: College FabLab / Makerspace (3D Printing Option)

- **Target Components:** 3 mm Shaft Eccentric Rotor Arm & ADXL345 Sensor Standoff Bracket.
- **Whom to Approach:** FabLab Student Coordinator, Makerspace Technician, or 3D Printing Operator.

- **What to Request & Exact CAD Specifications:**

Ask to print a small **Eccentric Vibration Cam/Arm** in standard PLA (20% to 30% infill, takes 8–12 minutes):

1. **Motor Shaft Bore:** A cylindrical hole of **3.0 mm** diameter with a flat 'D' profile (matching the N20 D-shaft flat side) so it slides tightly onto the shaft without slipping.
2. **Arm Length / Offset Radius (r):** Distance between the center of the shaft hole and the mass hole = **10 mm to 12 mm** .
3. **Off-Axis Bolt Hole:** A **3.2 mm** through-hole to pass the M3 bolt.
4. **Set Screw Hole (Optional but good):** A small 2.5 mm lateral hole on the side to tap an M3 grub screw if a friction fit isn't snug.

- **What to Say (Dialogue Script):**

> *"Hi, I have a tiny 5-gram 3D model for our microcontroller vibration rig. It's a small 3mm D-shaft eccentric cam arm to hold a small M3 bolt for vibration testing. Can we slice and print this on the Ender-3 / Prusa using PLA?"*

Location D: Local Nut-and-Bolt / Hardware Store (Cheruthuruthy / Shoranur / Thrissur)

- **Target Components:** M3 Machine Bolts, Washers, and Nyloc Locknuts.
- **Where to Go:** Any local retail hardware store, nut-bolt specialist, or motorcycle/bicycle spare parts shop.
- **What to Ask the Shopkeeper:**

> *"Bhai / Chetta, do you have small metric machine screws? I need:*

> • *One **M3 size, 15 mm length bolt** (Allen key or Star head).*

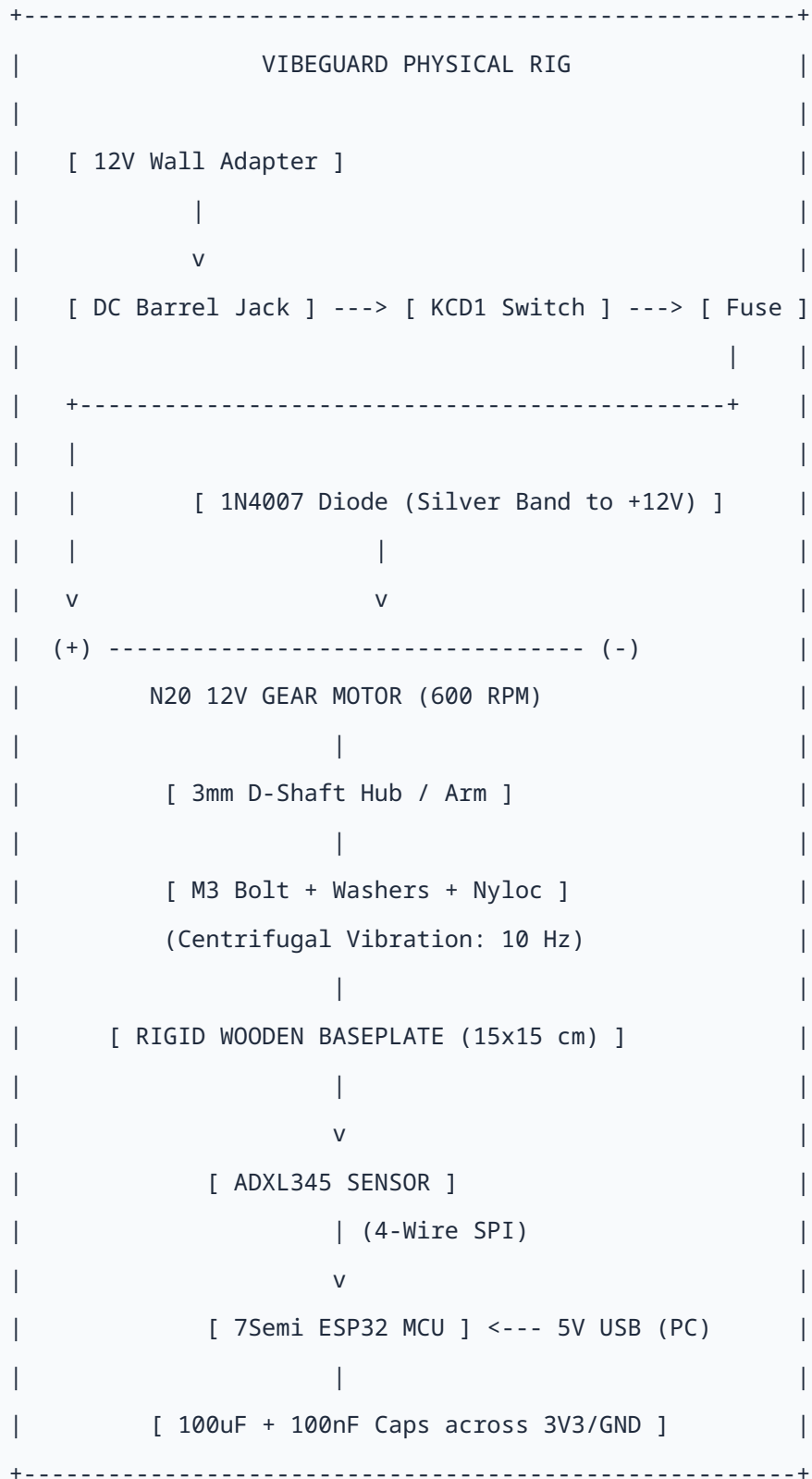
> • *Four **M3 flat steel washers**.*

> • *One **M3 Nyloc nut** (the nut with the white nylon plastic lock ring inside).*

> • *Two small **self-tapping wood screws** (around 10 mm length) to screw a bracket into wood."*

- **Cost:** Typically ₹5 to ₹10 for the entire handful.

4. Rig Assembly & Wiring Instructions: What to Do with Each Part



Step 1: Baseplate & Motor Mounting (G4-1 & G4-5)

- Place the flat wooden baseplate on a level desk.
- Position the metal N20 U-bracket roughly in the center-left of the board.
- Slide the N20 motor inside the bracket and tighten the bracket's clamping screw around the brass gearbox.
- Use the two small wood screws (**G4-5**) to fasten the bracket firmly into the wooden plank. Ensure the motor does not wiggle or tilt.

Step 2: Assembling the Eccentric Mass Rotor (G4-2, G4-3, G4-4, G4-6)

- Take the 3D-printed rotor arm (or 3 mm metal hub).
- Insert the **M3 × 15 mm bolt** through the off-axis hole.
- Thread **3 to 4 steel washers** onto the bolt. These washers provide the off-center unbalance mass ($m \approx 2.5 \text{ g}$).
- Screw on the **M3 Nyloc nut** and tighten firmly with a spanner/pliers.

> [!IMPORTANT]

> A standard plain nut can vibrate loose within 30 seconds of motor rotation. The **nylon ring inside the Nyloc nut** prevents it from ever vibrating loose.

- Push the completed hub/arm assembly onto the motor's 3 mm D-shaft and tighten the set screw (or check the tight friction fit).

Step 3: Rigid Sensor Mounting (ADXL345)

- Place the ADXL345 breakout board on the wooden base **15 mm to 25 mm away from the motor bearing face**.
- Fasten the sensor board firmly to the wood using two small screws or a rigid 3D-printed clip.

> [!CAUTION]

> **Prohibited:** Do NOT use squishy double-sided foam tape or hot glue. Soft materials act as vibration dampers/shock absorbers that attenuate high-frequency signals and distort FFT

readings.

Step 4: Soldering the 1N4007 Flyback Diode (G3-3)

- Inspect the 1N4007 diode: Notice the body has a **printed silver/white band** at one end. That end is the **Cathode**.
- Solder the diode directly across the two electrical solder lugs on the back of the N20 motor:

* **Cathode (Silver Band end):** Solder to the **Positive (+12V)** motor terminal.

* **Anode (Plain black end):** Solder to the **Negative (GND)** motor terminal.

- **Engineering Purpose:** An electric motor is a large inductive coil. When the KCD1 switch is turned OFF, the collapsing magnetic field creates a reverse voltage spike (back-EMF) that can exceed 100V. The 1N4007 instantly clamps this spike safely, protecting your switch contacts and nearby sensor logic.

Step 5: Wiring the DC Power Input Jack & Isolation

- Take the screw-terminal DC female jack:

* Connect the **+** terminal to one pin of the **KCD1 Rocker Switch**.

* From the other pin of the switch, wire to the **inline fuse holder** (carrying the 1A fuse).

* From the fuse holder, connect to the motor **Positive (+12V)** terminal.

* Connect the **-** terminal of the DC jack directly to the motor **Negative (GND)** terminal.

- **Absolute Galvanic Isolation Rule:**

* The 12V motor ground must **NEVER** touch the ESP32 ground pin.

* The ESP32 is powered exclusively via the 5V USB cable from the laptop.

* Testing rule: Touch multimeter probes between motor GND and ESP32 GND. The meter must read **Open Circuit / Infinite Ohms (∞ Ω)**.

Step 6: Power Decoupling on the Breadboard (G3-1, G3-2, G3-4)

- Seat the **7Semi ESP32-DEVKIT-E** firmly into the breadboard.

- **Install the 100 μ F Electrolytic Bulk Capacitor (G3-1):**

* Identify polarity: The side with a white/grey stripe and minus signs --- is the **Negative lead** (shorter pin). The other lead is **Positive**.

* Insert the **Positive lead** into the breadboard row connected to the ESP32 **3V3 pin (Pin 1)**.

* Insert the **Negative lead** into the breadboard **GND rail (Pin 14)**.

* Keep leads trimmed short ($\leq 5\text{ mm}$).

- **Install the 100 nF Ceramic Capacitor (G3-2):**

* Insert directly in parallel with the 100 μ F capacitor across the 3V3 and GND rows (ceramic caps are non-polarized).

* **Engineering Purpose:** When the ESP32 initiates Wi-Fi calibration or high-speed processing bursts, current spikes to 500 mA. The 100 μ F capacitor dumps stored charge in microseconds, preventing the AMS1117 LDO from dropping below the 2.70V Brownout threshold ($\tau_{st}: 0 \times 10$).

- **Install the 220 Ω Resistor (G3-4):**

*Insert in series with the **Red anode pin** of the RGB status LED connected to GPIO27. (Robu already supplied 330 Ω for Green/Blue).**

Step 7: Transparent Safety Guard (G4-7)

- Take a small, sturdy transparent container (clear acrylic box or thick clear plastic cup).
- Place it over the spinning eccentric hub and motor shaft.
- Fasten it to the wooden base with 2 small screws or brackets.
- **The 360° Clearance Test:** With power disconnected, reach in and turn the motor shaft by hand one full circle (360°). Ensure there is at least **5 mm of clear space** between the rotating bolt/washers and the inside walls of the guard.

5. Pre-Power Safety Sign-Off (Amith's Checklist)

Before plugging in the 12V wall adapter or the USB cable:

- **Locking Nut Checked:** M3 Nyloc nut is threaded past the nylon collar on the eccentric bolt.

- **Clearance Checked:** Manually rotated 360° — zero contact with baseplate or safety guard.
- **Flyback Diode Checked:** 1N4007 silver band is connected to the +12V motor terminal.
- **Galvanic Isolation Checked:** Multimeter confirms $\infty \Omega$ between 12V motor DC ground and ESP32 USB logic ground.
- **Bulk Cap Polarity Checked:** Negative stripe of 100 μF capacitor is connected to GND, not 3V3.
- **Fuse Verified:** 1A fast/time-delay cartridge fuse is inside the holder.